

Minimum Essential Characteristics (MEC)

High Throughput Bioreactor

The bioreactor shall meet the following MECs:

- Processes from up to 12 parallel, simultaneous cultures that can be used for support Design of Experiments (DOE).
Allows for the culture of both mammalian and microbial cells using available single use vessels with appropriate dual 20 mm pitched-blade or Rushton impeller with agitation speed ranging from 100 – 4500 rpm
- Provides tuning options including programmable Proportional, Integral, and Derivative (PID) control loops.
- Provides full control for pH and dissolved oxygen (DO) inclusive of control of gas to headspace and sparge and feeds.
- Provides an integrated sterile operating environment that allows for fully automated aseptic processing from batching to feeds and sample management with limited operator interaction.
- Allows connection to analytical instruments such as cell counter(s) and chemical/gas analyzers for automated sample processing and disposable DO, CO₂ and pH probes with automated feed and feedback controls and option for connection to exhaust gas analysis for Oxygen Uptake Rates (OUR)/Carbon dioxide Evolution Rates (CER)
- Allows for robotic liquid handler to sample from each bioreactor in volumes between 50 uL and 10 mL.
- Allows for robotic liquid handler to make liquid additions to bioreactors between 30 uL and 10 mL.
- Incorporates an automated pipette rinse with decontamination liquid.
- Allows for self-contained chiller for bioreactor cooling.
- Displays real-time data, store, and provide data in an off-line format and provide capability for external monitoring; software application enables predictive scale conversion.
- Provides for small volume high throughput with scalable features at a 100 - 250 mL working volume.
- Allows for scalability up to 30 L
- The system software and design shall be consistent with 21 CFR Part 11 requirements.
- Ensures the workstation (laptop) will keep working on battery and the controller to switch off in the event of power failure and without an external Uninterrupted Power Supply (UPS), with an option to connect to an external UPS to ensure an uninterrupted process.

Bioreactors and Process Control

- Vessel design shall be geometrically consistent with standard production scale bioreactors and fermenters and shall allow for scalable pitched blade (bioreactor) or Rushton (fermenter) impellers.
- Provides individual bioreactor controls including:
 - Individual temperature control with heating and cooling
 - Individual impeller speed control
 - Individual gas flow control
 - Individual pH testing and control
 - Individual sampling
- Allows for headspace gassing. Bioreactors shall be capable of simultaneous controlled addition of up to three gasses with mass flow sensor, either as sparge or overlay:
 - Mammalian
 - O₂
 - CO₂
 - N₂ | air
 - Microbial
 - O₂

- Air
- N2

- Capable of monitoring dissolved O₂ via an autoclavable or single-use probe.
- Capable of monitoring pCO₂ range from 15-200 mmHg
- Capable of monitoring DO range from 0-200%
- Capable of monitoring pH over the range of the process (2 – 8.5).
- Automatically monitors and controls the media temperature with an operational control range of 32-37°C and temperature probe shall be capable of monitoring between 0 - 55 °C.
- Allows for total gas or air flow up to 550 mL/min.
- Provides four positive displacement liquid pumps per bioreactor for high precision feed additions at low flow rates.
- Allows for integrated clean in place (CIP) | sterilize in place (SIP) for pumps and liquid lines.